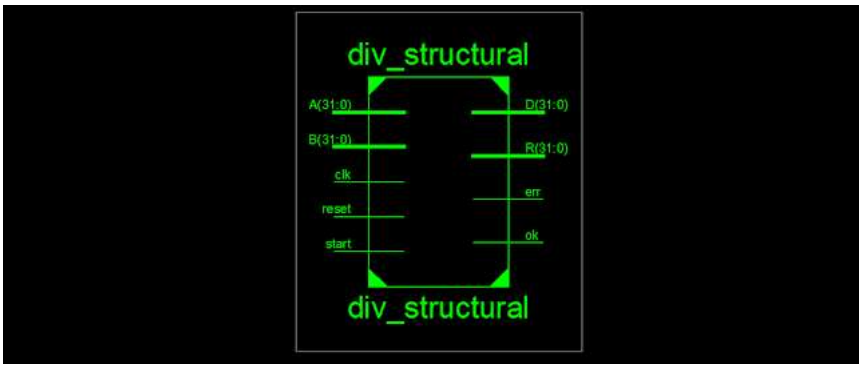
32-bit Unsigned Divider in Verilog

In this project, a 32-bit unsigned divider is implemented in Verilog using both structural and behavioral models. The Verilog code for the divider is synthesizable and can be implemented on FPGA.



Structural Verilog code for 32-bit unsigned divider:

```
`timescale 1ns / 1ns
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 32-bit divider
// Verilog code for divider using structural modelling
module div_structural(
  input      clk,
      input                    reset,
  input      start,
  input [31:0]  A,
  input [31:0]  B,
  output [31:0]  D,
  output [31:0]  R,
  output     ok ,   // =1 when ready to get the result
      output err
  );
  wire       active;   // True if the divider is running
  wire [4:0]    cycle;   // Number of cycles to go
      wire [4:0] cycle_d;
      wire [31:0]denom_d,work_d,result_d;
  wire [31:0]   result;   // Begin with A, end with D
  wire [31:0]   denom;   // B
  wire [31:0]   work;    // Running R
      wire start_n,clr,active_d;
  // Calculate the current digit
  wire [32:0]   sub = { work[30:0], result[31] } - denom;
      assign err = !B;
  // Send the results to our master
  assign D = result;
  assign R = work;
  assign ok = ~active;
      not n1(start_n,start);
      or o1(clr,reset,start_n);
      dff u1(.q(active), .d(active_d), .reset(clr), .clk(clk));
      assign active_d = active ? ((cycle==0)?1'b0:active): 1'b1;
      register_5 cycle_reg(cycle,cycle_d,1'b1,clr,clk);
      assign cycle_d = active?(cycle-5'd1):5'd31;
      register reg32_denom(denom,denom_d,1'b1,clr,clk);
      assign denom_d = active?denom:B;
      register reg32_work(work,work_d,1'b1,clr,clk);
      assign work_d = active?((sub[32]==0)?sub[31:0]:{work[30:0], result[31]}):0;
      register reg32_result(result,result_d,1'b1,clr,clk);
      assign result_d = active==1 ?((sub[32]==0)?{result[30:0], 1'b1}:{result[30:
```

```
endmodule
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 32-bit divider
// Verilog code for divider using structural modelling module
RegBit(BitOut, BitData, WriteEn,reset, clk);
output BitOut; // 1 bit of register
input BitData, WriteEn;
input reset,clk;
wire d,f1, f2; // input of D Flip-Flop
wire reset;
and #(50) U1(f1, BitOut, (~WriteEn));
and #(50) U2(f2, BitData, WriteEn);
or #(50) U3(d, f1, f2);
dff DFF0(BitOut, d, reset, clk);
endmodule
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 32-bit divider
// Verilog code for divider using structural modelling
module register_5(RegOut,RegIn,WriteEn,reset,clk);
output [4:0] RegOut;
input [4:0] RegIn;
input WriteEn,reset, clk;
RegBit     bit4 (RegOut[4], RegIn[4], WriteEn,reset,clk);
RegBit     bit3 (RegOut[3], RegIn[3], WriteEn,reset,clk);
RegBit     bit2 (RegOut[2], RegIn[2], WriteEn,reset,clk);
RegBit     bit1 (RegOut[1], RegIn[1], WriteEn,reset,clk);
RegBit     bit0 (RegOut[0], RegIn[0], WriteEn,reset,clk);
endmodule
// 32-bit register
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 32-bit divider
// Verilog code for divider using structural modelling
module register(RegOut,RegIn,WriteEn,reset,clk);
output [31:0] RegOut;
input [31:0] RegIn;
input WriteEn,reset, clk;
RegBit     bit31(RegOut[31],RegIn[31],WriteEn,reset,clk);
RegBit     bit30(RegOut[30],RegIn[30],WriteEn,reset,clk);
RegBit     bit29(RegOut[29],RegIn[29],WriteEn,reset,clk);
RegBit     bit28(RegOut[28],RegIn[28],WriteEn,reset,clk);
RegBit     bit27(RegOut[27],RegIn[27],WriteEn,reset,clk);
RegBit     bit26(RegOut[26],RegIn[26],WriteEn,reset,clk);
RegBit     bit25(RegOut[25],RegIn[25],WriteEn,reset,clk);
RegBit     bit24(RegOut[24],RegIn[24],WriteEn,reset,clk);
RegBit     bit23(RegOut[23],RegIn[23],WriteEn,reset,clk);
RegBit     bit22(RegOut[22],RegIn[22],WriteEn,reset,clk);
RegBit     bit21(RegOut[21],RegIn[21],WriteEn,reset,clk);
RegBit     bit20(RegOut[20],RegIn[20],WriteEn,reset,clk);
RegBit     bit19(RegOut[19],RegIn[19],WriteEn,reset,clk);
RegBit     bit18(RegOut[18],RegIn[18],WriteEn,reset,clk);
RegBit     bit17(RegOut[17],RegIn[17],WriteEn,reset,clk);
RegBit     bit16(RegOut[16],RegIn[16],WriteEn,reset,clk);
RegBit     bit15(RegOut[15],RegIn[15],WriteEn,reset,clk);
RegBit     bit14(RegOut[14],RegIn[14],WriteEn,reset,clk);
RegBit     bit13(RegOut[13],RegIn[13],WriteEn,reset,clk);
RegBit     bit12(RegOut[12],RegIn[12],WriteEn,reset,clk);
RegBit     bit11(RegOut[11],RegIn[11],WriteEn,reset,clk);
RegBit     bit10(RegOut[10],RegIn[10],WriteEn,reset,clk);
RegBit     bit9 (RegOut[9], RegIn[9], WriteEn,reset,clk);
RegBit     bit8 (RegOut[8], RegIn[8], WriteEn,reset,clk);
RegBit     bit7 (RegOut[7], RegIn[7], WriteEn,reset,clk);
RegBit     bit6 (RegOut[6], RegIn[6], WriteEn,reset,clk);
RegBit     bit5 (RegOut[5], RegIn[5], WriteEn,reset,clk);
```

```
RegBit     bit4 (RegOut[4], RegIn[4], WriteEn,reset,clk);
RegBit     bit3 (RegOut[3], RegIn[3], WriteEn,reset,clk);
RegBit     bit2 (RegOut[2], RegIn[2], WriteEn,reset,clk);
RegBit     bit1 (RegOut[1], RegIn[1], WriteEn,reset,clk);
RegBit     bit0 (RegOut[0], RegIn[0], WriteEn,reset,clk);
endmodule
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
module dff (q, d, reset, clk);
output q;
input d, reset, clk;
reg q; // Indicate that q is stateholding
always @(posedge clk or posedge reset)
if (reset)
q = 0;     // On reset, set to 0
else
q = d; // Otherwise out = d
endmodule
```

Behavioral Verilog code for 32-bit unsigned divider:

```
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 32-bit divider
// Verilog code for divider using behavioral modelling
module Divide(
  input      clk,
      input                    reset,
  input      start,
  input [31:0]  A,
  input [31:0]  B,
  output [31:0]  D,
  output [31:0]  R,
  output     ok ,   // =1 when ready to get the result
      output err
  );
  reg       active;   // True if the divider is running
  reg [4:0]    cycle;   // Number of cycles to go
  reg [31:0]   result;   // Begin with A, end with D
  reg [31:0]   denom;   // B
  reg [31:0]   work;    // Running R
  // Calculate the current digit
  wire [32:0]   sub = { work[30:0], result[31] } - denom;
      assign err = !B;
  // Send the results to our master
  assign D = result;
  assign R = work;
  assign ok = ~active;
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
  // The state machine
  always @(posedge clk,posedge reset) begin
    if (reset) begin
      active <= 0;
      cycle <= 0;
      result <= 0;
      denom <= 0;
      work <= 0;
    end
    else if(start) begin
      if (active) begin
        // Run an iteration of the divide.
        if (sub[32] == 0) begin
          work <= sub[31:0];
          result <= {result[30:0], 1'b1};
        end
        else begin
          work <= {work[30:0], result[31]};
          result <= {result[30:0], 1'b0};
        end
        if (cycle == 0) begin
          active <= 0;
        end
```

```
        cycle <= cycle - 5'd1;
      end
      else begin
        // Set up for an unsigned divide.
        cycle <= 5'd31;
        result <= A;
        denom <= B;
        work <= 32'b0;
        active <= 1;
      end
    end
  end
endmodule
```

Verilog testbench code for Structural divider:

```
`timescale 1ns / 1ns
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// Verilog project: Verilog code for 32-bit divider
// Testbench Verilog code for divider using structural modelling
module tb_div_structural;
     // Inputs
     reg clock;
     reg reset;
     reg start;
     reg [31:0] A;
     reg [31:0] B;
     // Outputs
     wire [31:0] D;
     wire [31:0] R;
     wire ok;
     wire err;
     // Instantiate the Unit Under Test (UUT)
     div_structural uut (
          .clk(clock),
          .start(start),
          .reset(reset),
          .A(A),
          .B(B),
          .D(D),
          .R(R),
          .ok(ok),
          .err(err)
     );
     initial begin
           clock = 0;
           forever #50 clock = ~clock;
     end
     initial begin
          // Initialize Inputs
          start = 0;
          A = 32'd1023;
          B = 32'd50;
          reset=1;
          // Wait 100 ns for global reset to finish
          #1000;
          reset=0;
    start = 1;
          #5000;
          $finish;
     end
endmodule
```

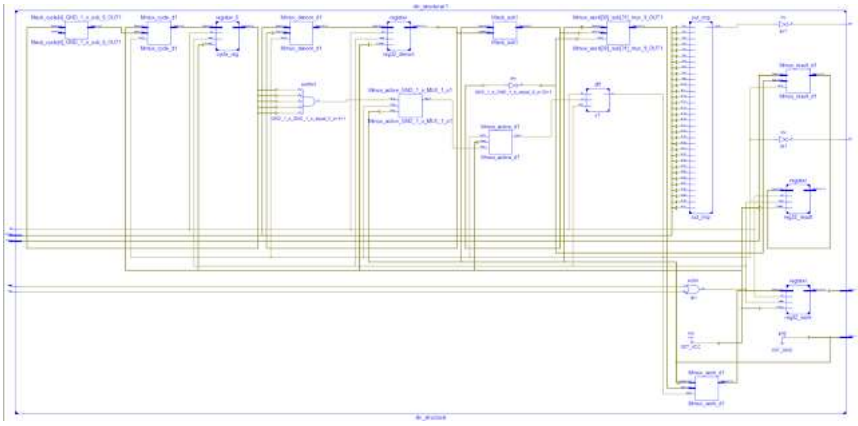
Verilog testbench code for Behavioral divider:

```
`timescale 1ns / 1ns
 // fpga4student.com FPGA projects, Verilog projects, VHDL projects
```

```
// Verilog project: Verilog code for 32-bit divider
// Testbench Verilog code for divider using behavioral modelling
module tb_divider;
     // Inputs
     reg clock;
     reg reset;
     reg start;
     reg [31:0] A;
     reg [31:0] B;
     // Outputs
     wire [31:0] D;
     wire [31:0] R;
     wire ok;
     wire err;
     // Instantiate the Unit Under Test (UUT)
     Divide uut (
          .clk(clock),
          .start(start),
          .reset(reset),
          .A(A),
          .B(B),
          .D(D),
          .R(R),
          .ok(ok),
          .err(err)
     );
     initial begin
           clock = 0;
           forever #50 clock = ~clock;
     end
     initial begin
          // Initialize Inputs
          start = 0;
          A = 32'd1023;
          B = 32'd50;
          reset=1;
          // Wait 100 ns for global reset to finish
          #1000;
          reset=0;
    start = 1;
          #5000;
          $finish;
     end
endmodule
```

Simulation waveform for the divider:

Synthesized RTL Schematic for the divider:



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